GEUNHYEONG LEE, Ph.D.

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RESEARCH VISION

AI-Driven Multiphysics Design and Optimization

My research focuses on AI-driven design and optimization for multiphysics engineering systems. I study how geometry representation, learning-based search, and physics-based verification can improve design discovery under computational and physical constraints. My background spans thermal-fluid, electromagnetic, and structural analysis; my current work includes BiSTO for turbulent cooling-channel topology optimization. My long-term goal is to develop reusable design methods across physical systems and connect computational optimization with manufacturing and experimental validation, with a focus on nuclear energy systems.

ACADEMIC APPOINTMENTS

Dec 2023 – Present
East Lansing, MI, US

Scientist (Staff) | Facility for Rare Isotope Beams (FRIB), Michigan State University
High Power Targetry Team

- Developed BiSTO (Binary Surrogate-based Topology Optimization), an AI-driven framework combining surrogate-guided topology search, solver verification, and geometric connectivity constraints for turbulent cooling-channel design.
- Lead the overall fabrication and installation effort for the FRIB Rotating Beam Dump, coordinating specialists in materials, systems engineering, thermomechanics, and radiation, with implementation targeted for 2028.
- Developed a hybrid genetic algorithm–reinforcement learning approach for FRIB beam-dump design optimization, increasing simulated average power-handling capability by 72% over the existing design.
- Developed a CNN–MLP surrogate for rotating beam-dump thermal limits (2.8% mean error against FEM in cross-validation; millisecond-scale inference).

Aug 2021 – Dec 2023
Daejeon, Korea

Postdoctoral Researcher | Korea Atomic Energy Research Institute (KAERI)
Applied Artificial Intelligence Lab

- Developed a deep reinforcement learning approach for heat-exchanger optimization, achieving 14.8% higher simulated heat transfer than conventional topology optimization; tested a metal-printed prototype.
- Designed a single-body, 3D-printable DC electromagnetic pump with 52% lower simulated current demand than a helical pump at matched pressure and flow rate.
- Designed flow channels for Samsung OLED susceptors using reinforcement learning.
- Applied AI anomaly detection to the KOMAC accelerator for operational stability.

EDUCATION

Mar 2016 – Aug 2021
Ulsan, Korea

Ph.D. in Nuclear Engineering (Combined M.S./Ph.D.) | UNIST
Ulsan National Institute of Science and Technology | Advisor: Hee Reyoung Kim
Thesis: Optimization of Lithium Charge Stripper through MHD Circulator Development

Mar 2010 – Feb 2016
Ulsan, Korea

B.S. in Nuclear Engineering & Mechanical Engineering (Dual Major) | UNIST

SCIE JOURNAL PUBLICATIONS

Published: 14 | First/co-first author†: 13 | Corresponding author: 5*

1. (2026) **Lee, G†***., Song, J., Quispe-Abad, R., Patil, M., & Kanemura, T. Optimization of the FRIB Beam Dump: A Hybrid Genetic Algorithm and Reinforcement Learning Approach. Nuclear Science and Techniques, 37, 184.
   Q1 · JCR Top 1.2% | IF 4.7 | DOI
2. (2026) **Lee, G†***. Conceptual Design of a 3D-Printable DC Electromagnetic Pump for Additive Manufacturing. Nuclear Engineering and Technology, 58(1), 103916.
   Q1 · JCR Top 13.1% | IF 2.9 | DOI
3. (2025) Kim, D.H†., Kim, H.S., Kwon, H.J., Yu, Y., & **Lee, G***. Deep learning-based anomaly detection in drift tube quadrupole operation for the KOMAC LINAC. Journal of Nuclear Science and Technology, 62(8), 709-714.
   Q1 · JCR Top 17.9% | IF 2.5 | DOI
4. (2024) **Lee, G†***., Joo, Y., Lee, S.U., Kim, T., Yu, Y., & Kim, H.G. Design optimization of heat exchanger using deep reinforcement learning. International Communications in Heat and Mass Transfer, 159, 107991.
   Q1 · JCR Top 6.1% | IF 6.4 | DOI
5. (2023) **Lee, G†***., Joo, Y†., Yu, Y., & Kim, H.G. Dual-fluid Topology Optimization of Printed-circuit Heat Exchanger with Low-pumping-power Design. Case Studies in Thermal Engineering, 49, 103318.
   Q1 · JCR Top 5.8% | IF 6.4 | DOI
6. (2023) Kang, T.U†., **Lee, G†**., & Kim, H.R*. Experimental characterization of the flowline of a lithium film formed using an electromagnetic thruster for a RAON prototype charge stripper.
   Nuclear Engineering and Design, 412, 112481.
   Q1 · JCR Top 21.2% | IF 1.9 | DOI
7. (2021) **Lee, G†**., & Kim, H.R*. Numerical study of Faraday-type nitrogen plasma magnetohydrodynamic generator. Journal of the Korean Physical Society, 78(7), 600-606. DOI
8. (2020) **Lee, G†**., & Kim, H.R*. The variable analysis of an MHD generator with electrical output of 10 kW for application to bi-plant method electricity generation.
   International Journal of Energy Research, 44(10), 8125-8132.
   Q1 · JCR Top 1.5% | IF 5.2 | DOI
9. (2019) **Lee, G†**., & Kim, H.R*. Mathematical approach for optimization of magnetohydrodynamic circulation system. Nuclear Engineering and Technology, 51(3), 654-664.
   Q1 · JCR Top 16.2% | IF 1.8 | DOI
10. (2018) **Lee, G†**., & Kim, H.R*. Magnetohydrodynamics approach for active decay heat removal system in future generation IV reactor. International Journal of Energy Research, 42(10), 3266-3278.
    Q1 · JCR Top 1.5% | IF 3.3 | DOI
11. (2018) **Lee, G†**., & Kim, H.R*. Numerical analysis of the electromagnetic force for design optimization of a rectangular direct current electromagnetic pump. Nuclear Engineering and Technology, 50(6), 869-876.
    Q1 · JCR Top 19.1% | IF 1.5 | DOI
12. (2018) **Lee, G†**., & Kim, H.R*. Magnetic-field analysis of an MHD channel in a liquid-metal circulation system of a prototype GenIV sodium fast reactor. Annals of Nuclear Energy, 115, 343-351. DOI
13. (2017) **Lee, G†**., & Kim, H.R*. Design analysis of DC electromagnetic pump for liquid sodium–CO2 reaction experimental characterization. Annals of Nuclear Energy, 109, 490-497.
    Q1 · JCR Top 13.6% | IF 1.5 | DOI
14. (2017) **Lee, G†**., & Kim, H.R*. Numerical investigation and comparison of the rectangular, cylindrical, and helical-type DC electromagnetic pumps. Magnetohydrodynamics, 53(2), 429-438. DOI

Under Review & Submitted

15. (Under review) **Lee, G†***., & Kanemura, T. A hybrid deep learning surrogate for thermal analysis of high-heat-flux devices. *Engineering Applications of Artificial Intelligence*. Preprint DOI
16. (Submitted) **Lee, G†***., & Kanemura, T. BiSTO: binary surrogate based topology optimization of turbulent cooling channels using deep learning.

OTHER JOURNAL PUBLICATIONS

1. (2025) **Lee, G**†*., Kim, T., Kim, Y., & Kim, G. Design and Analysis of an OLED Display Susceptor System Using Reinforcement Learning. Journal of Korean Applied Artificial Intelligence, 1(2), 1-6.

CONFERENCE PRESENTATIONS & PROCEEDINGS

*Completed: 34 | First/co-first author† | Corresponding author**

Completed Contributions

1. (2026) **Lee, G**†*., & Kanemura, T. BiSTO: AI Topology Optimization of Internal Cooling Channels for the FRIB High-Power Beam Dump toward Additive Manufacturing. Conference on Additive Manufacturing and Advanced Materials in Fusion (AM2F 2026), MIT Plasma Science and Fusion Center.
2. (2025) **Lee, G**†*., Bultman, N., Kanemura, T., Patil, M., Quispe-Abad, R., & Song, J. Design Improvement of a Minichannel Beam Dump Wing through AI-Driven Genetic Algorithms. 16th International Conference on Heavy Ion Accelerator Technology.
3. (2025) **Lee, G**†*., Song, J., Patil, M., Quispe-Abad, R., Bultman, N., & Kanemura, T. Design Optimization of the FRIB Beam Dump through Reinforcement Learning with Genetic Algorithm. 2025 American Nuclear Society Annual Conference.
4. (2025) Wei, J†*., et al. (incl. **Lee, G**.) FRIB Operations: First Three Years. 16th International Conference on Heavy Ion Accelerator Technology.
5. (2025) Song, J†*., Bultman, N., Cole, D., Jockheck, N., Kanemura, T., Larmann, M., Lee, D., **Lee, G**., Miller, S., Patil, M., Portillo, M., Quispe-Abad, R., Reaume, M., Simon, J., Steiner, M., & Wei, J. A Single-Slice Rotating Graphite Target at FRIB. 16th International Conference on Heavy Ion Accelerator Technology.
6. (2025) Kanemura, T†*., Abdelmegied, B., Bultman, N., Fletcher, E., Ginter, T., Iwai, R., Jockheck, N., LaVere, M., **Lee, G**., Marti, F., Maruta, T., Miller, S., Momozaki, Y., Ostroumov, P., Patil, M., Plastun, A., Quispe-Abad, R., Reaume, M., Simon, D., Song, J., Wakai, E., Wang, X., Wei, J., & Zhao, Q. High Power Targetry Devices at FRIB: Challenges, Status and Plan. 16th International Conference on Heavy Ion Accelerator Technology.
7. (2023) **Lee, G**†*., Yu, Y., Kim, H.G., & Kim, T. Application of Deep Learning Reinforcement Learning for Enhancement of Thermal Fluid Topology Optimization Performance. 3rd Applied Artificial Intelligence Conference.
8. (2023) **Lee, G**†*., Yu, Y., & Kim, H.G. Design of the Optimal Nuclear Heat Exchanger using Deep Reinforcement Learning. 10th Korea-China Workshop on Nuclear Reactor Thermal-Hydraulics.
9. (2023) Kim, D.H†*., **Lee, G**., Cho, S.Y., Lee, S.H., Kim, D.I., Kim, H.S., Kwon, H.J., & Yu, Y. Machine Learning-based Anomaly Detection in Drift Tube Quadrupole Operation for the KOMAC Linac. The 25th International Conference on Accelerators and Beam Utilizations.
10. (2023) **Lee, G**†*., Yu, Y., & Kim, H.G. Optimal Design of Nuclear Heat Exchanger Using 3D Printing-Based Deep Learning Reinforcement Learning. 2023 Korean Institute of Metals and Materials Fall Conference.
11. (2023) Lee, S.U†*., Kim, H.G., **Lee, G**., Kim, H., Yu, S., Oh, H., & Kim, Y. Development of Nuclear Component Manufacturing and Hybrid Fabrication Technology for Dissimilar Materials. 2023 Korean Institute of Metals and Materials Spring Conference.
12. (2023) **Lee, G**†*., & Kim, T. Formation of the LCD susceptor heat exchanger path using the deep reinforcement learning. 2nd Applied Artificial Intelligence Conference.
13. (2022) Jeon, B†., Ryu, S., Seo, H., Kim, M., Kim, S., **Lee, G**., Lim, H., & Yu, Y*. Development of Deep Learning-based Operator Support Systems for HANARO. 2022 HANARO Symposium.
14. (2022) **Lee, G**†*., Yu, Y., Kim, T., & Kim, S. Formation of a thermofluid temperature equalization flow path using reinforcement learning. 1st Applied Artificial Intelligence Conference.
15. (2022) **Lee, G**†*., Yu, Y., & Kim, H.G. Development of 3D Printing AI-Based Component Optimization Design Technology for Small Nuclear Reactor Materials. 2022 Korean Institute of Metals and Materials Spring Conference.

16. (2021) **Lee, G**†*., Joo, Y., Kim, H.G., & Yu, Y. The Analysis of Topology Optimized 3D Printing Heat Exchanger for the Small Nuclear Reactor. 2021 Korean Nuclear Society Autumn Meeting.
17. (2020) **Lee, G**†*., & Kim, H.R. The Electromagnetic Variable Analysis of 10 kW MHD Generator. 47th IEEE International Conference on Plasma Science.
18. (2020) **Lee, G**†*., & Kim, H.R. A Tendency of Electromagnetic Field Change in MHD Plasma Fluid Generator. 2020 Korean Physical Society Spring Meeting.
19. (2019) Kang, T.U†*., **Lee, G**., & Kim, H.R. Review Study on Reduction of NOx by Combining Thermal Power and Plasma MHD Power Generation. 2019 Korean Institute of Electrical Engineers Fall Conference.
20. (2019) **Lee, G**†*., Kang, T.U., & Kim, H.R. Simulation Analysis of the Characteristics of an Experimental MHD Generator. 2019 Korean Institute of Electrical Engineers Fall Conference.
21. (2019) **Lee, G**†., & Kim, H.R*. The Variable Optimization of MHD Generator with Electric Output of 10 kW. 11th PAMIR International Conference.
22. (2019) **Lee, G**†*., Kwak, J.S., & Kim, H.R. The Analysis of Experimental Magnetohydrodynamic Generator for Cogeneration Plant. 2019 Korean Institute of Electrical Engineers Spring Conference.
23. (2018) **Lee, G**†*., & Kim, H.R. Design Analysis of 10kW Magnetohydrodynamic Generator using Off-gas Plasma. 60th Annual Meeting of the APS Division of Plasma Physics.
24. (2018) Kang, T.U†*., **Lee, G**., & Kim, H.R. Analysis and Review of Basic Research on MHD Power Generation Technology. 2018 Korean Institute of Electrical Engineers Fall Conference.
25. (2018) **Lee, G**†*., & Kim, H.R. Characteristic Analysis of 10-kW Electric Output Faraday-type MHD Generator. 6th International Conference on Nuclear and Renewable Energy Resources.
26. (2018) **Lee, G**†*., & Kim, H.R. Finite Element Method Analysis of Rectangular-type DC Electromagnetic Pump for Active Heat Decay Removal System of Prototype Gen-IV Sodium Fast Reactor. 2018 Korean Physical Society Spring Meeting.
27. (2017) **Lee, G**†*., & Kim, H.R. The Comparative Analysis of Rectangular- and Helical-type DC Electromagnetic Pump for ADHRS in PGSFR. 2017 Korean Physical Society Autumn Meeting.
28. (2017) **Lee, G**†*., & Kim, H.R. The Optimization of Helical-Type DC Electromagnetic Pump for Active Decay Heat Removal System. Global 2017 International Nuclear Fuel Cycle Conference.
29. (2016) **Lee, G**†*., & Kim, H.R. The Pressure and Magnetic Flux Density Analysis of Helical-Type DC Electromagnetic Pump. 2016 Korean Nuclear Society Autumn Meeting.
30. (2016) **Lee, G**†*., & Kim, H.R. The Numerical Investigation and Comparison of the Rectangular, Cylindrical and Helical Type DC Electromagnetic Pumps. 10th PAMIR International Conference.
31. (2015) **Lee, G**†*., Kwak, J.S., & Kim, H.R. The Fundamental Design Analysis of the Rectangular and Helical Type-DC Conduction Electromagnetic Pump with the Flow Rate of 3 L/min. 23rd International Conference on Nuclear Engineering.
32. (2014) **Lee, G**†*., & Kim, H.R. The Design Analysis of the DC Electromagnetic Pump with the Flowrate of 3 L/min. 2014 American Nuclear Society Winter Meeting.
33. (2014) **Lee, G**†*., & Kim, H.R. The Optimum Design Analysis of the Small DC Electromagnetic Pump with Loop-Supported Type. 2014 Korean Nuclear Society Autumn Meeting.
34. (2014) **Lee, G**†*., & Kim, H.R. Preliminary Study for Radioactivity Evaluation of MSR compared with LWR. 2014 Korean Nuclear Society Spring Meeting.

PATENTS

1. (2021) Kim, H.R., Kang, T.U., & **Lee, G**. Thermal storage heat exchanger using liquid metal. Republic of Korea, Registration No. 10-2237876 / Application No. 10-2019-0171058
2. (2019) Kim, H.R., & **Lee, G**. Conductive material transfer device. Republic of Korea, Registration No. 10-1971612 / Application No. 10-2017-0075794

3. (2019) Kim, H.R., & **Lee, G**. Conductive material transfer device. Republic of Korea, Registration No. 10-2017117 / Application No. 10-2017-0063580
4. (2018) Kim, H.R., & **Lee, G**. Conductive material transfer device. Republic of Korea, Registration No. 10-1860895 / Application No. 10-2017-0025724
5. (2018) Kim, H.R., & **Lee, G**. Conductive material transfer device. Republic of Korea, Registration No. 10-1860898 / Application No. 10-2017-0015538
6. (2018) Kim, H.R., & **Lee, G**. Electromagnetic pump. Republic of Korea, Registration No. 10-1908115 / Application No. 10-2016-0177052

TEACHING

Mentoring

(2026) Adam Sypitkowski, Michigan State University, Accelerator Science and Engineering Traineeship (ASET) program at FRIB. Mentored in AI-driven geometry optimization for accelerator simulations.

(2026) Undergraduate and graduate students, Michigan State University. Advised on AI applications in research, including wedge structural optimization, and on experimental methods for measuring target-system thermal conductance.

Teaching Experience

(2023) Lecturer, Foundation of AI. International School on Beam Dynamics and Accelerator Technology.

(2016) Instructor, Nuclear Comprehensive Design Project. UNIST.

Teaching Assistant

(2021) Liquid Metal Magnetohydrodynamics I. UNIST.

(2020) Liquid Metal Magnetohydrodynamics II. UNIST.

(2019) Special Lecture on Nuclear Engineering (MHD electrical power generation). UNIST.

(2017–2019) Fundamental of Electromagnetics. UNIST.

ACHIEVEMENTS

Awards

(2019) Doosan Heavy Industries & Construction Research Scholarship. Korean Nuclear Society.

(2016) Excellent Poster Award. Korean Nuclear Society, Korea.

(2015) First Award of Excellence for MNE Research Internship. UNIST.

Invited Talks

(2026) AI-Driven Design Optimization for FRIB Beam Intercepting Devices. KAERI.

(2023) The Optimal Nuclear Heat Exchanger Design through Reinforcement Learning. BrainLink X-lab.

(2023) Optimal design of 3D printing-based heat exchanger using reinforcement learning. UNIST.

(2023) Optimal design of nuclear power plant heat exchanger using 3D printing-based deep learning reinforcement learning. 2023 Korean Institute of Metals and Materials Fall Meeting.

(2022) Optimal design analysis of major reactor parts using AI technology. UNIST.

(2022) Development of Small Nuclear Reactor Main Part, 3D printing AI-based Optimization Design Technology. 2022 Korean Institute of Metals and Materials Spring Meeting.

Software Registration

(2022) **Lee, G**., Yu, Y., & Kim, T. Heat exchanger path generation program through reinforcement learning. Republic of Korea, Registration No. C-2022-049451.

Certification

(2021) Certified AI Expert, Korean Standards Association, No. C-2021-0004668.

SERVICE

Journal Peer Review (32 reviews)

Journal	Reviews	Period
Energy	6	2024–2026
International Communications in Heat and Mass Transfer	9	2025–2026
Annals of Nuclear Energy	4	2025–2026
Electronic Research Archive	9	2025–2026
Journal of Thermal Analysis and Calorimetry	1	2026
Journal of the Korean Physical Society	1	2023
Energy Engineering	2	2024

Leadership

Student Representative, Department of Nuclear Engineering, UNIST.

SELECTED RESEARCH PROJECT EXPERIENCE

2023–2026
FRIB

AIP Mini-channel and Water Beam Dump
Developed a simulation-based optimization framework for beam-dump design, evaluated thermal performance, and contributed to engineering reviews of beam-dump hardware.

2023–2026
FRIB

AIP Post-target Shielding
Led thermal design reviews and high-power system thermal simulation for operation

2023–2026
FRIB

Accelerator Operation Labor
Thermal device simulation, optimization, validation, and operational support

2022
KAERI

Technical Support for AI-Based Improvement of Temperature Uniformity in Next-Generation Display Manufacturing Systems
Applied reinforcement learning to optimize flow-channel layouts for display susceptors.

2021–2023
KAERI

Development of Original 3D Printing Technology for Manufacturing Components of Small Nuclear Reactor Materials
Developed deep reinforcement learning models for heat-exchanger design; contributed to topology-optimization analysis and experimental validation of a metal-printed prototype.

2021–2023
KAERI

Establishment of Intelligent Operation Platform for HANARO and Research Facilities
Anomaly detection in the accelerator system using AI

2021–2022
Hyundai

Development of Deep Learning Solver for Reducing Iterations in Topology Optimization
Supported topology-optimization simulations for deep-learning solver development.

2020–2021
Institute for Basic Science

Research on the Development of Prototype Lithium Charge Stripper
Design optimization of an electromagnetic pump and liquid lithium film formation system

2018–2021
KEPCO Research Institute

Plasma Generation Technology of High Temperature and Electric Production Technology of 10 kW Magnetic Generator
Investigation of MHD generator technology and design analysis

2016–2018 KAERI	**Design and Performance Evaluation about EM Pump for Test of Coolant Loop of Prototype Gen-IV Sodium-cooled Fast Reactor** Feasibility analysis and performance evaluation of DC electromagnetic pumps for SFR
2016 Institute for Basic Science	**Development of Prototype DC Electromagnetic Pump** Design optimization of a DC electromagnetic pump